

Semester	B.E. Semester VII – Computer Engineering
Subject	R Programming
Subject Professor In- charge	Prof. Sanjeev Dwivedi
Assisting Teachers	Prof. Sanjeev Dwivedi
Laboratory	M-411B

Student Name	Sneha Gadhari
Roll Number	23102B0055
Grade and Subject Teacher's Signature	

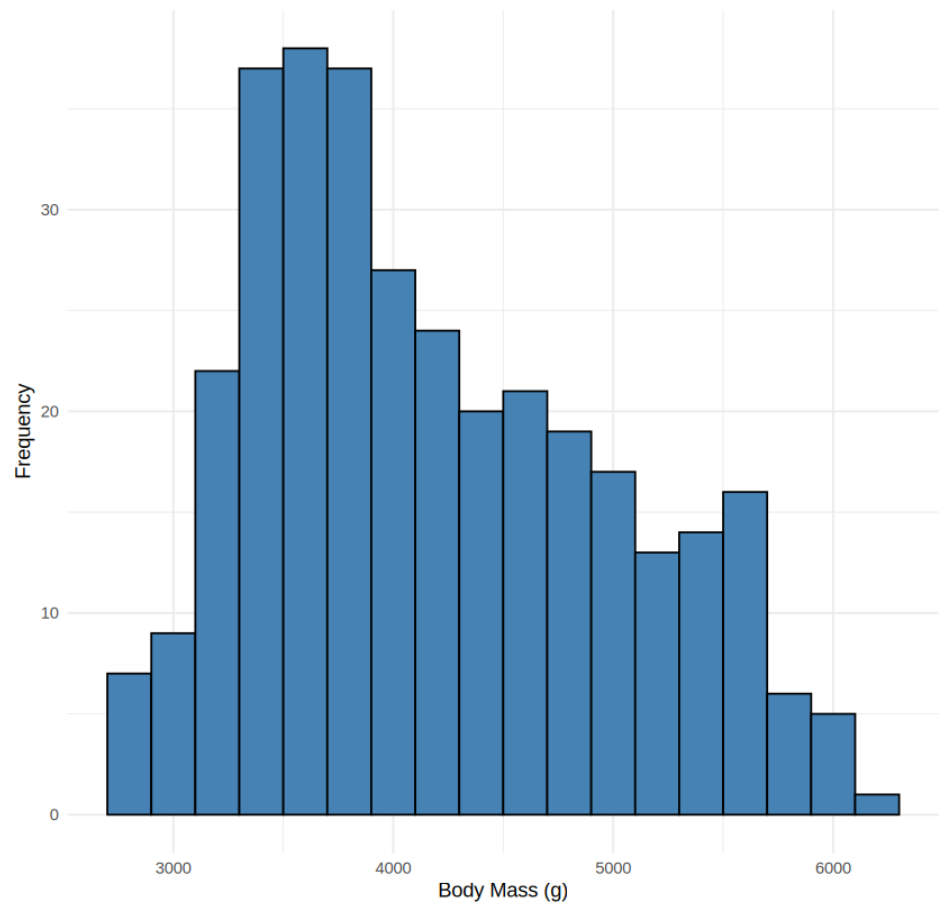
Experiment Number	06
Experiment Title	Statistical Analysis of Physical Characteristics of Palmer Penguins

Resources / Apparatus Required	Hardware: Computer system	Software: Colab
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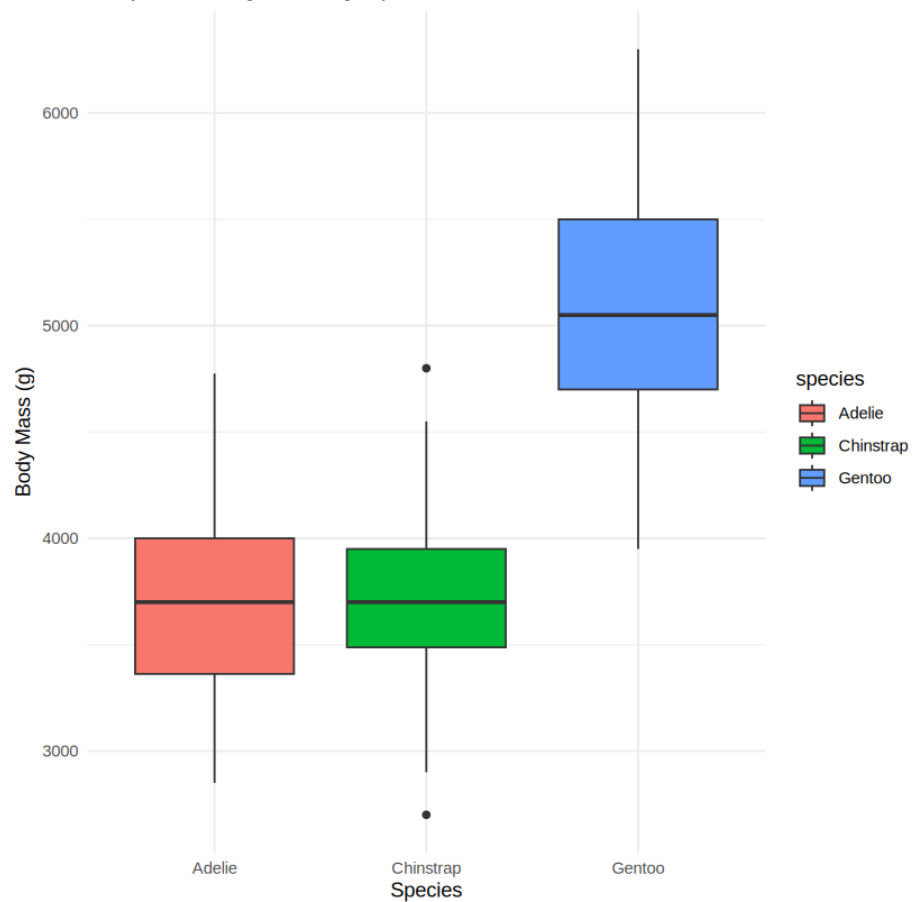
Objective	The objective of this experiment is to perform a comprehensive statistical analysis of the Palmer Penguins dataset to examine whether physical characteristics primarily body mass and flipper length differ significantly across penguin species (Adélie, Chinstrap, Gentoo) and sex. The analysis combines descriptive statistics, hypothesis testing, effect-size estimation, ANOVA, post-hoc testing, and non-parametric alternatives to draw statistically sound and well-interpreted conclusions.
Description	This experiment uses the Palmer Penguins dataset (344 observations, loaded directly via URL) to carry out an end-to-end statistical workflow in R. Data Preparation - Dataset loaded from a public URL (no manual download required) and missing values removed. - Key variables used: species, sex, body_mass_g, flipper_length_mm. Descriptive Statistics - Computed mean, median, min, max, variance, standard deviation, Q1, Q3, IQR, skewness, and kurtosis for body mass both overall and species-wise. - Distributions visualized using histograms, boxplots, and density plots. Hypothesis Testing (Sex-based comparison) - Stated null (no difference) and alternative (significant difference) hypotheses for mean body mass between male and female penguins. - Verified normality assumption using the Shapiro-Wilk test and QQ-plots. - Conducted an independent two-sample t-test, computed a 95% confidence interval for the mean difference, and calculated Cohen's d for effect size. One-Way ANOVA (Species comparison) - Checked ANOVA assumptions: normality (Shapiro-Wilk, QQ-plots) and homogeneity of variance (Levene's test). - Performed one-way ANOVA (body_mass_g ~ species) and reported the F-statistic, degrees of freedom, and p-value. - Applied Tukey's HSD post-hoc test to identify which species pairs differ significantly. Non-Parametric Analysis - Performed the Kruskal-Wallis test as a robustness check

	and compared its conclusions against the ANOVA results. Two-Way ANOVA - Examined the combined effects of species, sex, and their interaction on body mass (body_mass_g ~ species * sex). Additional Analysis (Flipper Length) - Repeated the normality, homogeneity, ANOVA, Tukey HSD, and Kruskal-Wallis workflow for flipper_length_mm and compared findings against the body mass results. Visualization - Species-wise and sex-wise boxplots, density plots, QQ-plots, Tukey HSD interval plots, and a combined boxplot-with-jitter comparison plot were generated with proper titles and axis labels throughout.																																																																																																																																				
Output	<pre>'data.frame': 333 obs. of 8 variables: \$ species : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ... \$ island : chr "Torgersen" "Torgersen" "Torgersen" "Torgersen" ... \$ bill_length_mm : num 39.1 39.5 40.3 36.7 39.3 38.9 39.2 41.1 38.6 34.6 ... \$ bill_depth_mm : num 18.7 17.4 18 19.3 20.6 17.8 19.6 17.6 21.2 21.1 ... \$ flipper_length_mm: int 181 186 195 193 190 181 195 182 191 198 ... \$ body_mass_g : int 3750 3800 3250 3450 3650 3625 4675 3200 3800 4400 ... \$ sex : Factor w/ 2 levels "female","male": 2 1 1 1 2 1 2 1 2 2 ... \$ year : int 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ... - attr(*, "na.action")= 'omit' Named int [1:11] 4 9 10 11 12 48 179 219 257 269 attr(*, "names")= chr [1:11] "4" "9" "10" "11" ... A data.frame: 6 x 8</pre> <table><tr><th></th><th>species</th><th>island</th><th>bill_length_mm</th><th>bill_depth_mm</th><th>flipper_length_mm</th><th>body_mass_g</th><th>sex</th><th>year</th></tr><tr><th></th><th><fct></th><th><chr></th><th><dbl></th><th><dbl></th><th><int></th><th><int></th><th><fct></th><th><int></th></tr><tr><td>1</td><td>Adelie</td><td>Torgersen</td><td>39.1</td><td>18.7</td><td>181</td><td>3750</td><td>male</td><td>2007</td></tr><tr><td>2</td><td>Adelie</td><td>Torgersen</td><td>39.5</td><td>17.4</td><td>186</td><td>3800</td><td>female</td><td>2007</td></tr><tr><td>3</td><td>Adelie</td><td>Torgersen</td><td>40.3</td><td>18.0</td><td>195</td><td>3250</td><td>female</td><td>2007</td></tr><tr><td>5</td><td>Adelie</td><td>Torgersen</td><td>36.7</td><td>19.3</td><td>193</td><td>3450</td><td>female</td><td>2007</td></tr><tr><td>6</td><td>Adelie</td><td>Torgersen</td><td>39.3</td><td>20.6</td><td>190</td><td>3650</td><td>male</td><td>2007</td></tr><tr><td>7</td><td>Adelie</td><td>Torgersen</td><td>38.9</td><td>17.8</td><td>181</td><td>3625</td><td>female</td><td>2007</td></tr></table> <pre>A tibble: 3 x 12</pre> <table><tr><th>species</th><th>Mean</th><th>Median</th><th>Min</th><th>Max</th><th>Variance</th><th>SD</th><th>Q1</th><th>Q3</th><th>IQR</th><th>Skewness</th><th>Kurtosis</th></tr><tr><th><fct></th><th><dbl></th><th><dbl></th><th><int></th><th><int></th><th><dbl></th><th><dbl></th><th><dbl></th><th><dbl></th><th><dbl></th><th><dbl></th><th><dbl></th></tr><tr><td>Adelie</td><td>3706.164</td><td>3700</td><td>2850</td><td>4775</td><td>210332.4</td><td>458.6201</td><td>3362.5</td><td>4000</td><td>637.5</td><td>0.2770077</td><td>2.411479</td></tr><tr><td>Chinstrap</td><td>3733.088</td><td>3700</td><td>2700</td><td>4800</td><td>147713.5</td><td>384.3351</td><td>3487.5</td><td>3950</td><td>462.5</td><td>0.2419413</td><td>3.463681</td></tr><tr><td>Gentoo</td><td>5092.437</td><td>5050</td><td>3950</td><td>6300</td><td>251478.3</td><td>501.4762</td><td>4700.0</td><td>5500</td><td>800.0</td><td>0.0449965</td><td>2.242850</td></tr></table>		species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year		<fct>	<chr>	<dbl>	<dbl>	<int>	<int>	<fct>	<int>	1	Adelie	Torgersen	39.1	18.7	181	3750	male	2007	2	Adelie	Torgersen	39.5	17.4	186	3800	female	2007	3	Adelie	Torgersen	40.3	18.0	195	3250	female	2007	5	Adelie	Torgersen	36.7	19.3	193	3450	female	2007	6	Adelie	Torgersen	39.3	20.6	190	3650	male	2007	7	Adelie	Torgersen	38.9	17.8	181	3625	female	2007	species	Mean	Median	Min	Max	Variance	SD	Q1	Q3	IQR	Skewness	Kurtosis	<fct>	<dbl>	<dbl>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	Adelie	3706.164	3700	2850	4775	210332.4	458.6201	3362.5	4000	637.5	0.2770077	2.411479	Chinstrap	3733.088	3700	2700	4800	147713.5	384.3351	3487.5	3950	462.5	0.2419413	3.463681	Gentoo	5092.437	5050	3950	6300	251478.3	501.4762	4700.0	5500	800.0	0.0449965	2.242850
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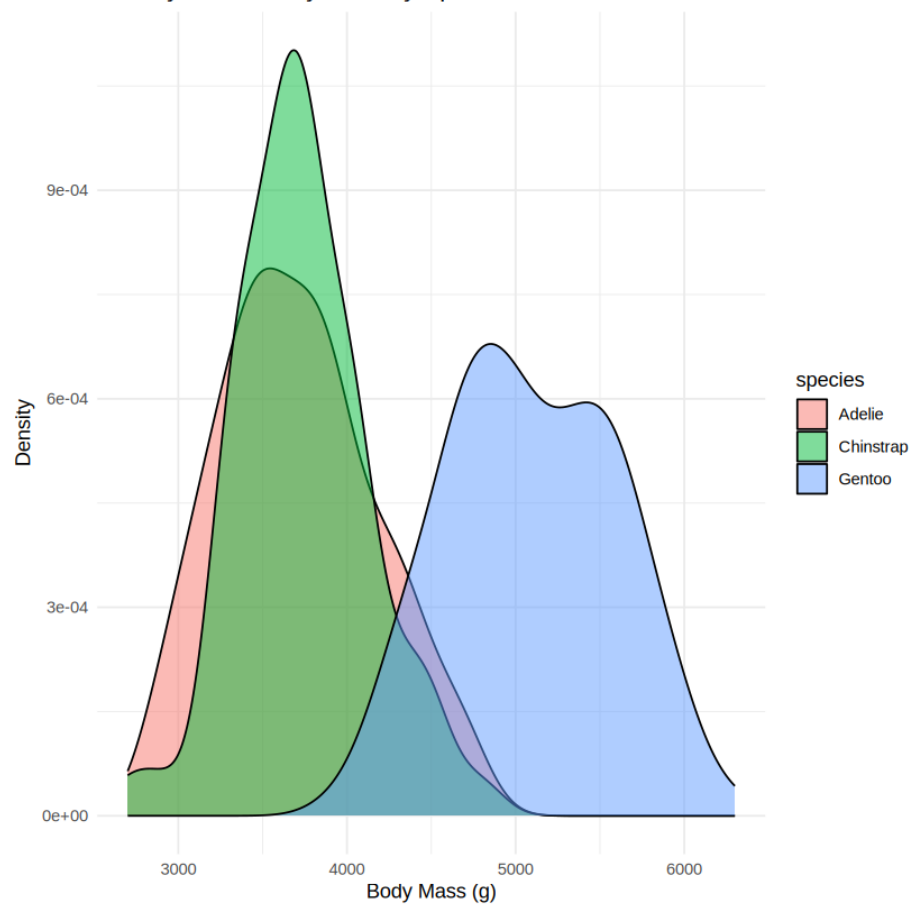
Histogram of Penguin Body Mass



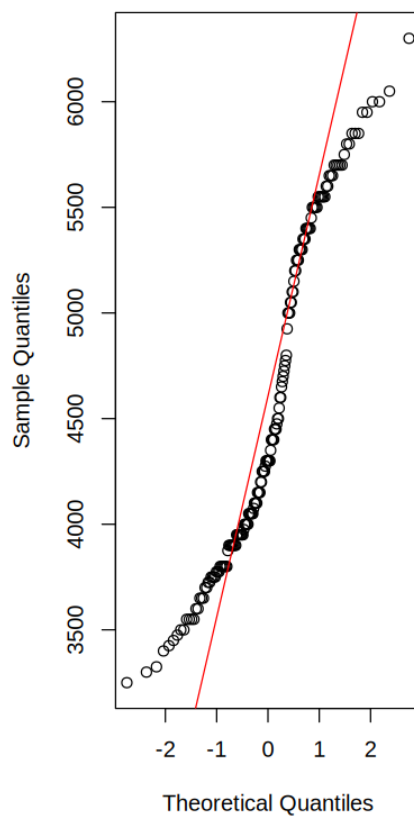
Boxplot of Body Mass by Species



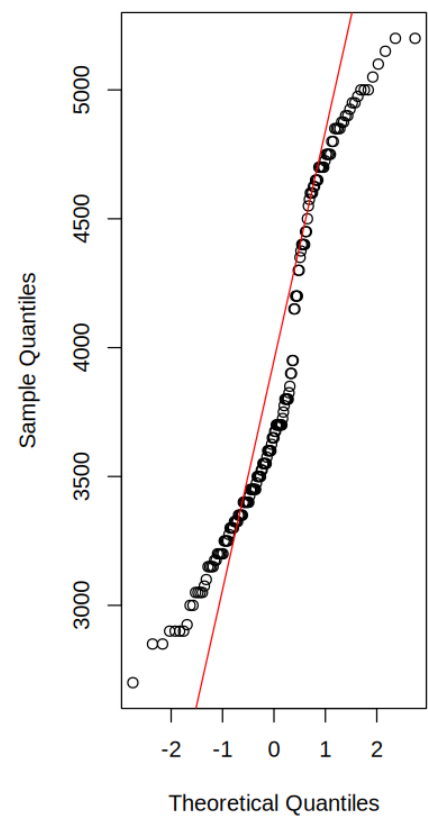
Density Plot of Body Mass by Species



QQ-Plot: Male Body Mass



QQ-Plot: Female Body Mass



```
data: body_mass_g by sex
t = -8.5545, df = 323.9, p-value = 4.794e-16
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
-840.5783 -526.2453
sample estimates:
mean in group female    mean in group male
      3862.273           4545.685
```

\$Adelie

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2850	3362	3700	3706	4000	4775

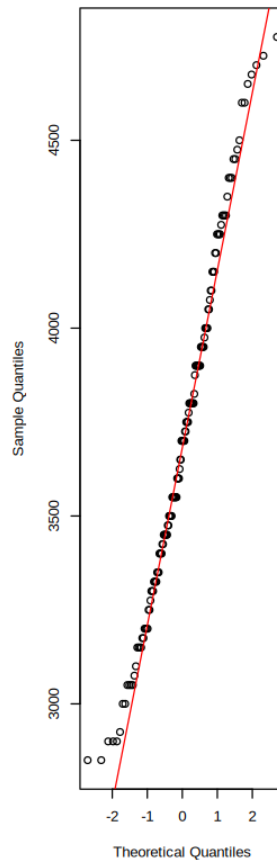
\$Chinstrap

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2700	3488	3700	3733	3950	4800

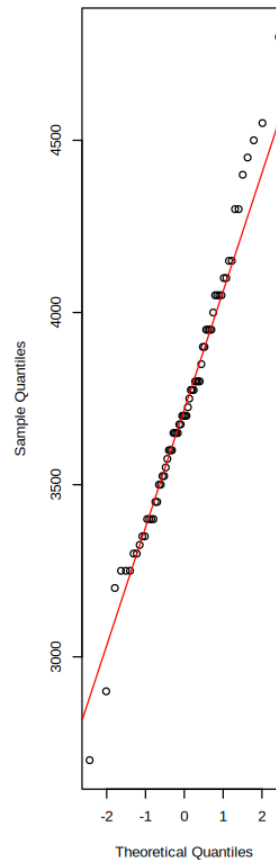
\$Gentoo

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
3950	4700	5050	5092	5500	6300

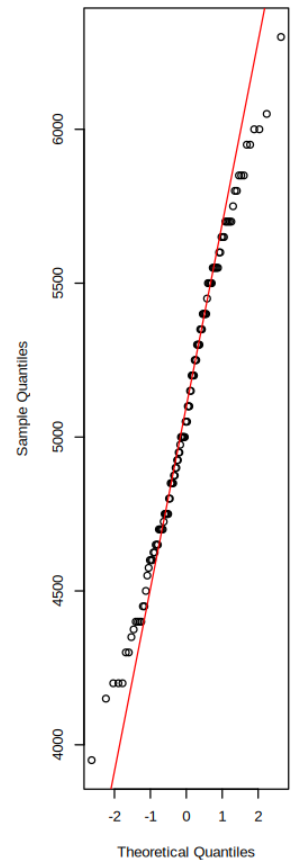
QQ-Plot: Adelie



QQ-Plot: Chinstrap



QQ-Plot: Gentoo



```
$Adelie
```

```
Shapiro-Wilk normality test
```

```
data: X[[i]]
```

```
W = 0.98116, p-value = 0.04232
```

```
$Chinstrap
```

```
Shapiro-Wilk normality test
```

```
data: X[[i]]
```

```
W = 0.98449, p-value = 0.5605
```

```
$Gentoo
```

```
Shapiro-Wilk normality test
```

```
data: X[[i]]
```

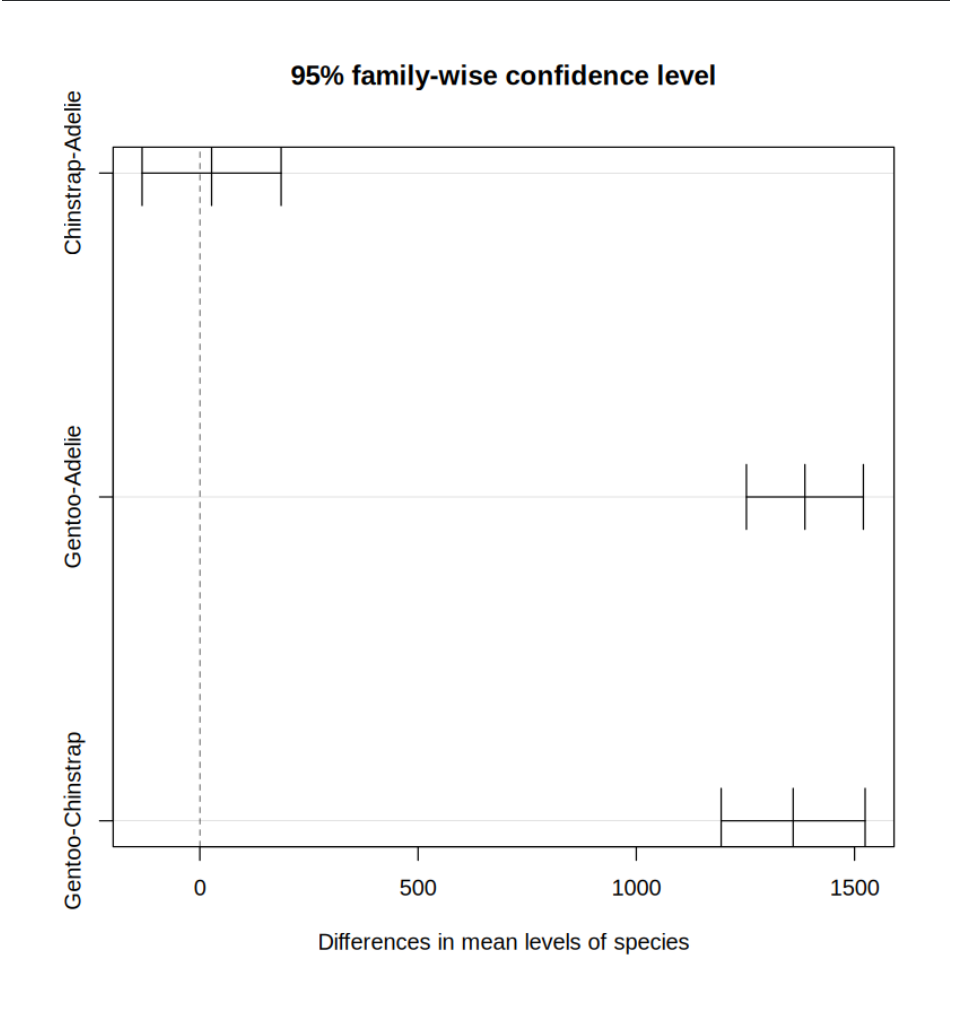
```
W = 0.98606, p-value = 0.2605
```

```
A anova: 2 × 3
```

	Df	F value	Pr(>F)
	<int>	<dbl>	<dbl>
group	2	5.134899	0.006367054
	330	NA	NA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	145190219	72595110	341.9	<2e-16 ***
Residuals	330	70069447	212332		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

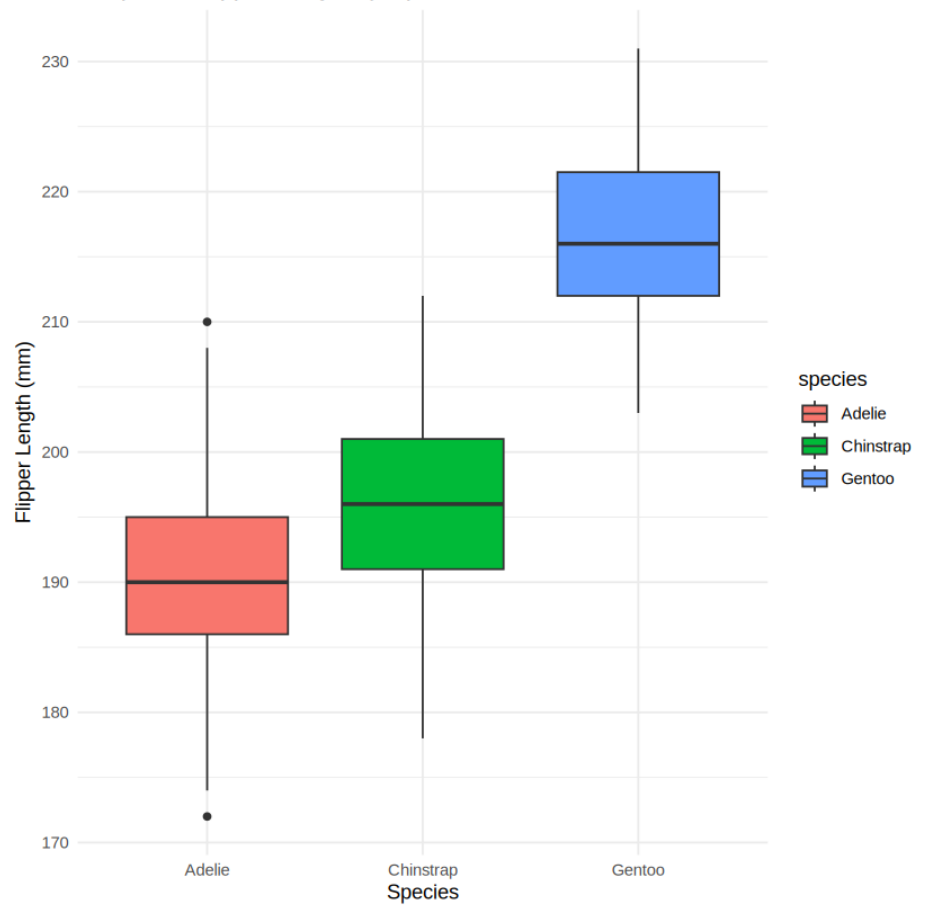


Kruskal-Wallis rank sum test	
data: body_mass_g by species	
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16	

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	145190219	72595110	758.358	< 2e-16 ***
sex	1	37090262	37090262	387.460	< 2e-16 ***
species:sex	2	1676557	838278	8.757	0.000197 ***
Residuals	327	31302628	95727		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

Boxplot of Flipper Length by Species



Boxplot of Body Mass by Sex

